

# Estimating The Size Distribution of California Wildfires

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## 1 Introduction

In the last few years wildfires in California have grown in size and destructiveness to the environment. (Western) In this paper we estimate the distribution of wildfire sizes in California by utilizing data from the Cal Fire Hub, a database supplied by the California Department of Forestry and Fire Protection. The dataset has been narrowed down to include only wildfires with a size of 50,000 acres or more, comprising of 45 wildfires from the years 2017-2022, all of which occurred in California. (California) Please refer to Figure 1 for a histogram illustrating the size distribution of our dataset.

After determining an appropriate approximation (Pareto) of the size distribution of California wildfires, we estimate the annual count of wildfires in the state that are 300,000 acres or larger. We hope that this information is valuable for state bureaucrats to plan and allocate resources effectively. While we acknowledge that decisions about infrastructure require additional considerations beyond wildfire size, we believe our analysis contributes a crucial aspect to the overall understanding of wildfire management.

The structure of our paper is as follows: section two describes the statistical methods we employ to evaluate the distribution of the size of California wildfires. Section three provides our results. Section four briefly discusses those results.

## 2 Methods

For our analysis, we operated under the assumption that the size distribution of California wildfires, denoted  $Y$  in this paper, follows a Pareto distribution. Figure 1 below provides our data histogram, and if this Pareto assumption is to be believed, it would be a necessary and sufficient condition that the transformed distribution  $X = \ln(Y/y_m)$  would follow an exponential distribution with the same parameter value. The value of  $y_m$  here is the minimum value of the distribution: 50,000 for the wildfire dataset. Figure 2 provides a Q-Q plot comparing our data to the theoretical exponential distribution.<sup>1</sup> With this, we find that the

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<sup>1</sup>We use exponential because R does not have a method for comparing our data to the theoretical Pareto distribution. Additionally we use the shape parameter  $\alpha$  from our maximum likelihood estimate. We discuss

data reasonably follows a Pareto distribution ( $Y \sim \text{Pareto}(\alpha)$ ).<sup>2</sup> The left tail of our data is marginally heavier than the theoretical distribution and we do not have much data on the far right tail to get a clear picture of how well the data fits the theoretical distribution on the far right tail. Still, the data seems to fit the theoretic distribution well enough to use the Pareto distribution to generate our estimates. Additionally, a Pareto distribution seems reasonable, as we anticipate smaller wildfires are more common than larger ones. Our objective is to estimate  $\alpha$ , the shape parameter determining the slope of the Pareto distribution.

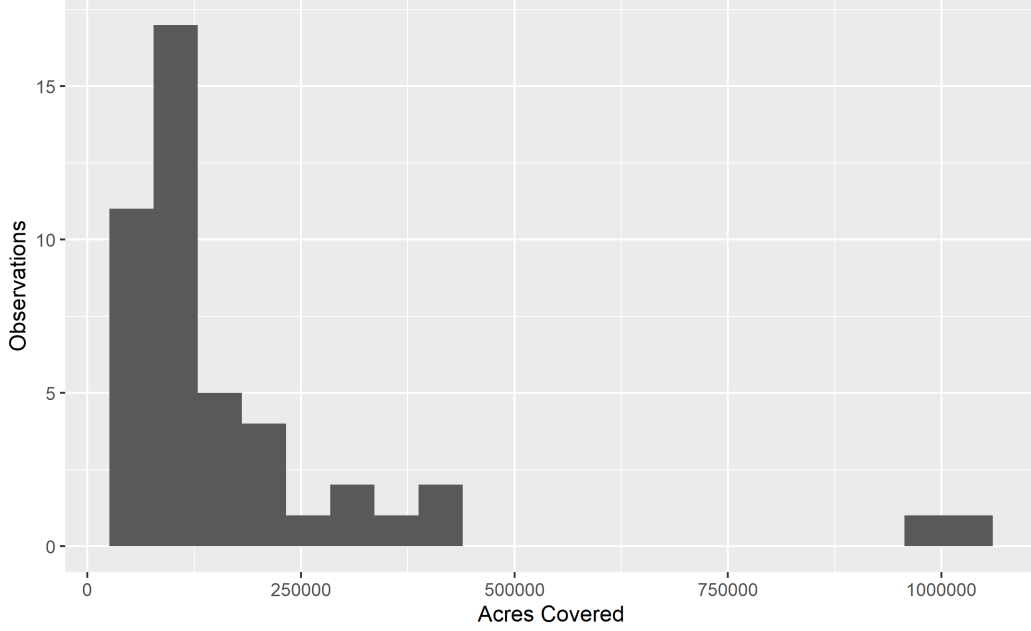


Figure 1: Histogram of Fires Data

As previously stated, given  $Y \sim \text{Pareto}(\alpha)$ , it follows that  $\ln(Y/y_m) \sim \text{Exp}(\alpha)$ . Consequently, we establish that  $1/\hat{\alpha}_{mle} \sim \text{Gamma}(n, n\alpha)$ , and  $\hat{\alpha}_{mom} = \frac{\bar{Y}}{\bar{Y} - y_m}$ , where  $\hat{\alpha}_{mle}$  and  $\hat{\alpha}_{mom}$  are our respective maximum likelihood and method of moments estimators. To generate a point estimate for  $\alpha$  we consider both a method of moments (MoM) and maximum likelihood estimations (MLE). To construct a 95% confidence interval for  $\alpha$  we consider four typical methods: exact pivotal, formula t, bootstrap MLE, and bootstrap MoM. Letting  $qGn(a)$  denote the  $a$ -th quantile of the  $\text{Gamma}(n, n)$  distribution, we have the exact pivotal interval:

$$\hat{\alpha}_{mle} \cdot qGn(0.0025) < \alpha < \hat{\alpha}_{mle} \cdot qGn(0.975).$$

Letting  $qt(a)$  denote the  $a$ -th quantile of the  $t_{n-1}$  distribution,  $\bar{X}$  be the sample mean of  $X = \ln(Y/y_m)$ , and  $s$  be the sample standard deviation of  $X$ , we have the formula t statistic interval:

$$\frac{1}{\bar{X} - qt(0.025)s/\sqrt{n}} < \alpha < \frac{1}{\bar{X} - qt(0.975)s/\sqrt{n}}.$$

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this in the results section.

<sup>2</sup>All statements in this paragraph are proven in the appendix.

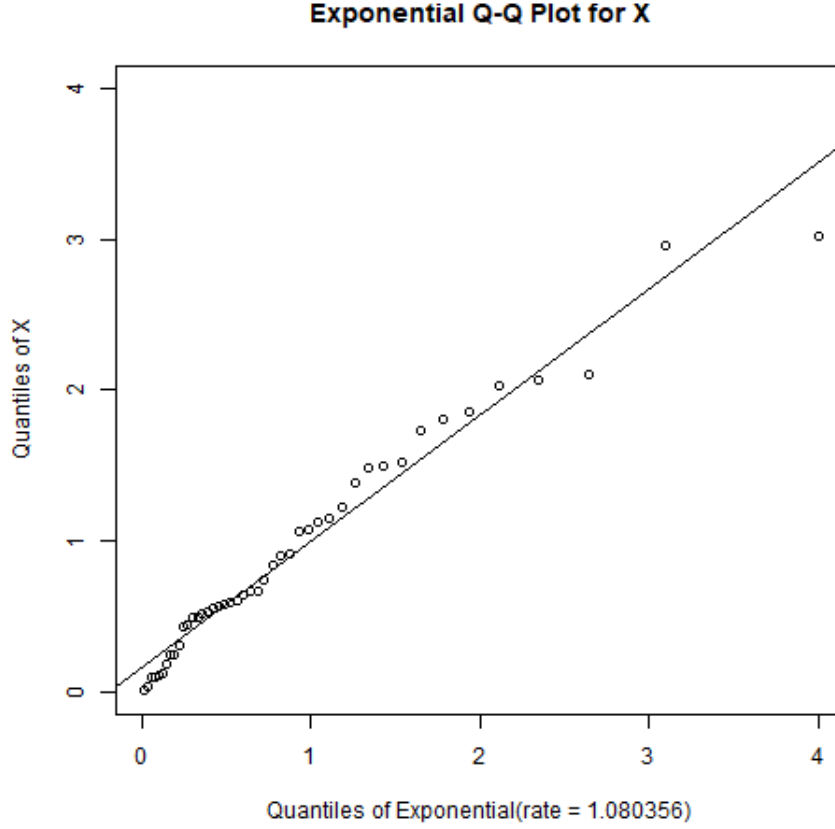


Figure 2: Q-Q Plot of Transformed Fire Data (X)

Letting  $b_{mle}(a)$  and  $b_{mom}(a)$  denote the  $a$ -th quantiles of the bootstrap  $\hat{\alpha}_{mle}$  and bootstrap  $\hat{\alpha}_{mom}$  distributions, we have the bootstrap MLE and bootstrap MoM intervals:

$$b_{mle}(0.025) < \alpha < b_{mle}(0.975),$$

$$b_{mom}(0.025) < \alpha < b_{mom}(0.975).$$

The results of all confidence interval methods are further explored in the next section. As an extension, we also consider the bootstrap t method for constructing a confidence interval. Letting  $b_t(a)$  denote the  $a$ -th quantile of the bootstrap t distribution, using the  $t$  statistic for  $X$ , we have the bootstrap t interval:

$$\frac{1}{\bar{X} - b_t(0.025)s/\sqrt{n}} < \alpha < \frac{1}{\bar{X} - b_t(0.975)s/\sqrt{n}}$$

The bootstrap t method requires fewer observations and does not require us to make strong assumptions about the underlying distribution so it is worth exploring in addition to our other methods.<sup>3</sup>

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<sup>3</sup>The second point is true of the bootstrap t method broadly, but in our case, to construct the t-statistic we still have to assume an underlying Pareto distribution

To determine a preferred method for a point estimate of the shape parameter  $\alpha$ , we compare the root mean squared error (RMSE) of the MLE and MoM estimators, seeking to minimize this error across varying true  $\alpha$  levels and sample sizes  $n$ . To determine a preferred method for constructing a confidence interval we vary both the sample size  $n$  and the true shape parameter  $\alpha$  to then measure both the simulated coverage rate and average simulated width of the CI's. We seek the method that primarily maximizes the coverage rate and secondarily minimizes the confidence interval width across the varying settings.

### 3 Results

Running our simulations to determine estimator RMSE, CI coverage rate, and CI width for  $n = 15, 50, 100, 100$  and  $\alpha = 0.5, 1.5, 2.5$ , we have a collection of tables summarizing these results, respectively Table 1, Table 4, and Table 10.

In broad strokes, we may observe that both increasing the sample size and the shape parameter increases the coverage rate for all methods of CI construction as seen in Table 4. For the method of moments estimator, the maximum coverage rate is 92% while for all other estimators the maximum coverage rate is 95%. Notably the exact pivotal statistic method and the bootstrap t method achieve a 95% coverage rate with smaller  $n$  and  $\alpha$  with great consistency, compared to all other methods.

With respect to our point estimates, we see that increasing  $\alpha$  tends to increase the RMSE of both estimators with the exception of the  $\alpha = 0.5$  values for the MoM estimator. Furthermore, increasing  $n$  consistently decreases the RMSE of both estimates, which is in-line with how the standard error of the distributions in question shrink as  $n$  increases. The least preferred method for constructing a point estimate is the method of moments. Note that the maximum likelihood estimator has lower RMSE than the method of moments estimator for all values for  $n$  and  $\alpha$  but the method of moments estimator is particularly poor when  $\alpha = 0.5$ . When  $0 < \alpha < 1$ ,  $E[Y] = \infty$ , meaning the method of moments cannot be used to give an accurate estimate by its reliance on a finite expectation. This is supported in Table 5 with the zero coverage rate of the bootstrap MoM method when  $\alpha < 1$ . This fact, in addition to the consistently smaller RMSE of the MLE estimate over all  $\alpha$  and  $n$  as seen in Table 1, leads us to choose to use the maximum likelihood estimator to generate a point estimate for  $\alpha$ . Our point estimate given by the maximum likelihood estimation of  $\alpha$  is then 1.08. Figure 2 provides a comparison of our data to the theoretical Pareto distribution using shape parameter  $\alpha = 1.08$ . Indeed, we see that our data fits the theoretic distribution well.

With respect to the CI methods, we note that the coverage for the bootstrap MoM method is incredibly poor, the bootstrap MLE and formula t methods performed moderately well, and the exact pivotal and bootstrap t methods performed near perfectly. The best methods for determining a confidence interval for  $\alpha$  are the exact pivotal statistic and the bootstrap t method.<sup>4</sup> For these methods the quality of our confidence interval, as measured by the coverage rate, is largely independent of the  $n$  or  $\alpha$  as seen in Table 4. We desire a CI that is more consistently meeting its coverage rate over one that is narrower, so we only consider

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<sup>4</sup>Note that the exact pivotal method is strictly preferred to the MLE

Table 1: RMSE Comparison Between MLE and MOM Estimators for Varying  $\alpha$  and  $n$

|          |     | $n$  |      |      |      |
|----------|-----|------|------|------|------|
|          |     | 15   | 50   | 100  | 1000 |
| $\alpha$ | 0.5 | 0.15 | 0.08 | 0.05 | 0.02 |
|          | 1.5 | 0.46 | 0.23 | 0.16 | 0.05 |
|          | 2.5 | 0.73 | 0.37 | 0.26 | 0.08 |

Table 2: MLE Estimator

|          |     | $n$  |      |      |      |
|----------|-----|------|------|------|------|
|          |     | 15   | 50   | 100  | 1000 |
| $\alpha$ | 0.5 | 0.55 | 0.51 | 0.51 | 0.50 |
|          | 1.5 | 0.52 | 0.28 | 0.21 | 0.09 |
|          | 2.5 | 0.76 | 0.40 | 0.27 | 0.10 |

Table 3: MoM Estimator

Table 4: Coverage Rate of True  $\alpha$  for Varying Methods of 95% CI and  $n$

|          |     | $n$  |      |      |      |
|----------|-----|------|------|------|------|
|          |     | 15   | 50   | 100  | 1000 |
| $\alpha$ | 0.5 | 0.00 | 0.00 | 0.00 | 0.00 |
|          | 1.5 | 0.64 | 0.67 | 0.73 | 0.78 |
|          | 2.5 | 0.82 | 0.88 | 0.91 | 0.92 |

Table 5: Bootstrap MoM Method

|          |     | $n$  |      |      |      |
|----------|-----|------|------|------|------|
|          |     | 15   | 50   | 100  | 1000 |
| $\alpha$ | 0.5 | 0.89 | 0.92 | 0.95 | 0.94 |
|          | 1.5 | 0.89 | 0.92 | 0.94 | 0.95 |
|          | 2.5 | 0.90 | 0.94 | 0.93 | 0.94 |

Table 6: Bootstrap MLE Method

|          |     | $n$  |      |      |      |
|----------|-----|------|------|------|------|
|          |     | 15   | 50   | 100  | 1000 |
| $\alpha$ | 0.5 | 0.91 | 0.93 | 0.95 | 0.95 |
|          | 1.5 | 0.91 | 0.94 | 0.94 | 0.95 |
|          | 2.5 | 0.92 | 0.94 | 0.94 | 0.95 |

Table 7: Formula T Method

|          |     | $n$  |      |      |      |
|----------|-----|------|------|------|------|
|          |     | 15   | 50   | 100  | 1000 |
| $\alpha$ | 0.5 | 0.95 | 0.95 | 0.95 | 0.94 |
|          | 1.5 | 0.95 | 0.94 | 0.95 | 0.94 |
|          | 2.5 | 0.96 | 0.96 | 0.95 | 0.95 |

Table 8: Exact Pivotal Method

|          |     | $n$  |      |      |      |
|----------|-----|------|------|------|------|
|          |     | 15   | 50   | 100  | 1000 |
| $\alpha$ | 0.5 | 0.94 | 0.94 | 0.95 | 0.94 |
|          | 1.5 | 0.95 | 0.95 | 0.95 | 0.95 |
|          | 2.5 | 0.95 | 0.95 | 0.95 | 0.95 |

Table 9: Bootstrap T Method

the widths of the CI's after finding the methods of best coverage. Because the exact pivotal statistic method on average has a narrower width as seen in Table 10, we chose to use it when constructing our confidence interval. Using the exact pivotal statistic method we are 95% confident that the true shape parameter  $\alpha$  for the distribution of California wildfires sizes is between 0.788 and 1.418.

We can estimate the expected count of California wildfires per year which are larger than

Table 10: 95% CI Average Width for Varying Methods,  $\alpha$ , and  $n$ 

|          |     | $n$   |       |       |       |
|----------|-----|-------|-------|-------|-------|
|          |     | 15    | 50    | 100   | 1000  |
| $\alpha$ | 0.5 | 0.261 | 0.071 | 0.034 | 0.003 |
|          | 1.5 | 1.506 | 0.764 | 0.554 | 0.244 |
|          | 2.5 | 2.652 | 1.379 | 0.992 | 0.353 |

Table 11: Bootstrap MoM Method

|          |     | $n$   |       |       |       |
|----------|-----|-------|-------|-------|-------|
|          |     | 15    | 50    | 100   | 1000  |
| $\alpha$ | 0.5 | 0.573 | 0.286 | 0.197 | 0.062 |
|          | 1.5 | 1.726 | 0.861 | 0.591 | 0.185 |
|          | 2.5 | 2.838 | 1.432 | 0.996 | 0.309 |

Table 12: Bootstrap MLE Method

|          |     | $n$   |       |       |       |
|----------|-----|-------|-------|-------|-------|
|          |     | 15    | 50    | 100   | 1000  |
| $\alpha$ | 0.5 | 0.797 | 0.312 | 0.205 | 0.062 |
|          | 1.5 | 1.954 | 0.938 | 0.617 | 0.187 |
|          | 2.5 | 4.497 | 1.557 | 1.041 | 0.311 |

Table 13: Formula T Method

|          |     | $n$   |       |       |       |
|----------|-----|-------|-------|-------|-------|
|          |     | 15    | 50    | 100   | 1000  |
| $\alpha$ | 0.5 | 0.539 | 0.282 | 0.197 | 0.062 |
|          | 1.5 | 1.611 | 0.848 | 0.592 | 0.186 |
|          | 2.5 | 2.658 | 1.419 | 0.989 | 0.310 |

Table 14: Exact Pivotal Method

|          |     | $n$   |       |       |       |
|----------|-----|-------|-------|-------|-------|
|          |     | 15    | 50    | 100   | 1000  |
| $\alpha$ | 0.5 | 0.640 | 0.294 | 0.199 | 0.062 |
|          | 1.5 | 1.931 | 0.887 | 0.600 | 0.186 |
|          | 2.5 | 3.167 | 1.473 | 1.011 | 0.310 |

Table 15: Bootstrap T Method

300,000 acres as follows:

$$\begin{aligned}
& E(\# \text{ California wildfires} \geq 300,000 \text{ acres per year}) \\
&= E(\# \text{ California wildfires per year} \geq 50,000 \text{ acres}) \times P(\text{area} \geq 300,000 \mid \text{area} \geq 50,000) \\
&= E(\# \text{ California wildfires per year} \geq 50,000 \text{ acres}) \times (1 - F(300,000)) \\
&= 9 \times \left( \frac{50,000}{300,000} \right)^\alpha
\end{aligned}$$

We estimate the number of wildfires greater than 50,000 acres a year to be 9 as our dataset spans 5 years and contains 45 fires of that size or larger. Therefore our point estimate for the expected count of California wildfires over 300,000 acres is 1.3 wildfires per year, by setting  $\alpha$  to our point estimate of 1.08. We can further transform our 95% CI of  $\alpha$  into a 95% CI for the number of wildfires covering more than 300,000 acres per year by setting  $\alpha$  to be the endpoints of the  $\alpha$  CI. Thus, we are 95% confident that the true number of California wildfires larger than 300,000 acres per year is between 0.71 and 2.19 wildfires per year. Given that our dataset contains 45 wildfires over 5 years, or roughly nine per year, we find that just shy of 15% of California which that are larger than 50,000 acres have a size that over 300,000 acres.

## 4 Discussion

In this paper we estimated the number of California wildfires larger than 300,000 acres to be 1.3. To estimate this, we worked under the assumption that the distribution of California wildfires' sizes (in acres) is well approximated by a Pareto distribution, and we constructed several point estimates and confidence intervals for a Pareto distribution's shape parameter. After considering the coverage rates and confidence interval widths of the various methods across a variety of initial conditions, we found that the exact pivotal method was the preferred method for constructing a confidence interval. By comparing the RMSE between the method of moments and maximum likelihood estimators, we found the MLE to have consistently better performance and thus was our preferred point estimate method. Our point estimate for  $\alpha$ , the shape parameter of the Pareto distribution approximating the size distribution of California wildfires, is 1.08 and our 95% confidence interval is (0.788, 1.418). Consequently, our point estimate for the expected number of California wildfires per year larger than 300,000 acres is 1.3 and our 95% confidence interval is (0.71, 2.19).

Notably, 0 is not in our confidence interval for the expected number of sufficiently large wildfires a year. We can then reasonably expect one, and potentially two California wildfires over 300,000 acres each year. This result suggests that the state of California should be prepared for a wildfire of such size—they happen every year.

Our analysis can be improved by considering a longer time interval—including a larger number of wildfires. Additionally, the analysis will benefit from longitudinal evaluations of how the size and distribution of California wildfires have changed over time. A more robust analysis must include all wildfires—not just those that are over 50,000 acres in size. Our analysis was also limited because we considered the entire state of California which is vast and therefore likely has large heterogeneity in climate. As for the limitations on our simulations, we were limited by the computational power of our devices, which if improved, could allow us to develop larger bootstrap distribution and use a larger number of iterations of the simulation as a whole.

## 5 Appendix

Here we offer an explanation of key results we reference in our paper.

If  $Y \sim \text{Pareto}(y_m, \alpha)$  then  $\ln(Y/y_m) \sim \text{Exp}(\alpha)$ .

We will proceed by generating the CDF for  $\ln(Y/y_m)$ . Recall that the CDF models  $P(\ln(Y/y_m) \leq z)$  where  $z$  is some value of the support. Note that

$$P(\ln(\frac{Y}{y_m}) \leq z) = P(Y \leq y_m \cdot e^z).$$

We know that  $Y \sim \text{Pareto}(y_m, \alpha)$  so using the CDF for  $\text{Pareto}(y_m, \alpha)$  yields

$$P(Y \leq y_m \cdot e^z) = 1 - \left( \frac{y_m}{y_m \cdot e^z} \right)^\alpha = 1 - e^{-z\alpha}.$$

Note that by the CDF for the Pareto distribution  $e^z y_m > y_m$  or  $e^z > 1$  which implies that  $z > 0$ . Using the form of the CDF for an exponential distribution we conclude that  $\ln(Y/y_m) \sim \text{Exp}(\alpha)$ , as desired.

The maximum likelihood estimate for the Pareto parameter  $\alpha$  is  $\hat{\alpha}_{mle} = \frac{1}{\frac{1}{n} \sum_{i=1}^n \ln(Y_i/y_m)}$ .

We first note that our likelihood function is

$$\prod_{i=1}^n \frac{\alpha y_m^\alpha}{Y_i^{\alpha+1}}.$$

Note that our likelihood function can be 0 if  $\min(Y_i) < y_m$ . Our log likelihood function is

$$\sum_{i=1}^n \frac{1}{\alpha} + \alpha \cdot \ln(y_m) - (\alpha + 1) \cdot \ln(Y_i).$$

Taking the derivation with respect to  $\alpha$  and setting it equal to zero yields

$$\sum_{i=1}^n \frac{1}{\alpha} + \ln(y_m) - \ln(Y_i) = 0.$$

Solving algebraically, we conclude that

$$\hat{\alpha}_{mle} = \frac{1}{\frac{1}{n} \sum_{i=1}^n \ln\left(\frac{Y_i}{y_m}\right)}$$

as desired.

The reciprocal of the MLE estimator is given by:  $\frac{1}{\hat{\alpha}_{mle}} \sim \text{Gamma}(n, n\alpha)$ .

Note that our solution from part b can be rewritten as

$$\frac{1}{\hat{\alpha}_{mle}} = \frac{1}{n} \sum_{i=1}^n \ln\left(\frac{Y_i}{y_m}\right).$$

We have shown in part a that

$$\ln\left(\frac{Y_i}{y_m}\right) \sim \text{Exp}(\alpha).$$

Recall that the sum of i.i.d exponential random variables follow a gamma distribution. Therefore

$$\sum_{i=1}^n \ln \left( \frac{Y_i}{y_m} \right) \sim \text{Gamma}(n, \alpha).$$

By properties of the gamma distribution,

$$\frac{1}{n} \sum_{i=1}^n \ln \left( \frac{Y_i}{y_m} \right) \sim \text{Gamma}(n, n\alpha).$$

Thus we conclude

$$\frac{1}{\hat{\alpha}_{mle}} \sim \text{Gamma}(n, n\alpha)$$

as desired.

*Derivation of an exact 95% confidence interval for  $\alpha$  using the pivotal method:*

By the work above, we note that  $\frac{\alpha}{\hat{\alpha}_{mle}} \sim \text{Gamma}(n, n)$ . Let  $qGn(x)$  denote the  $x$ -th quantile of the  $\text{Gamma}(n, n)$  distribution. To determine an exact 95% CI for  $\alpha$ , we see that

$$P \left( qGn(0.025) < \frac{\alpha}{\hat{\alpha}_{mle}} < qGn(0.975) \right) = 0.95.$$

The inner equality is equivalent to

$$\hat{\alpha}_{mle} \cdot qGn(0.025) < \alpha < \hat{\alpha}_{mle} \cdot qGn(0.975),$$

which are our respective upper and lower bounds for the exact 95% CI.

*Deriving an approximate 95% confidence interval for  $\alpha$  using the  $t_{n-1}$  distribution:*

Let  $qt(x)$  denote the  $x$ -th quantile of the  $t_{n-1}$  distribution. To determine an approximate 95% CI for  $\alpha$  using this distribution, we note that for  $X_i = \ln(Y_i/y_m)$  and  $s$  as the sample standard deviation of the  $X_i$ :

$$\frac{1}{\hat{\alpha}_{mle}} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Further, for large  $n$ :

$$\frac{\bar{X} - \frac{1}{\alpha}}{s/\sqrt{n}} \stackrel{\text{approx}}{\sim} t_{n-1}.$$

Thus, we see again for large  $n$  the following should hold:

$$P \left( qt(0.025) < \frac{\bar{X} - \frac{1}{\alpha}}{s/\sqrt{n}} < qt(0.975) \right) = 0.95.$$

This inner inequality is then equivalent to

$$\frac{1}{\bar{X} - qt(0.025)s/\sqrt{n}} < \alpha < \frac{1}{\bar{X} - qt(0.975)s/\sqrt{n}},$$

which are our respective upper and lower bounds for the approximate 95% CI.

*Deriving the method of moments estimator:*

To find the method of moments estimator, we desire to solve the following equation for  $\hat{\alpha}$ , as  $\alpha > 1$ :

$$\frac{1}{n} \sum_{i=1}^n Y_i = \bar{Y} = E_{\hat{\alpha}}[Y] = \frac{\hat{\alpha}y_m}{\hat{\alpha} - 1}.$$

This gives

$$\hat{\alpha}\bar{Y} - \bar{Y} = \hat{\alpha}y_m, \quad \text{and further,}$$

$$\hat{\alpha}_{mom} = \frac{\bar{Y}}{\bar{Y} - y_m}.$$

Thus, we have the method of moments estimator for  $\alpha$ .

## 6 R Markdown Appendix

See the attached pdf for the full R Markdown file detailing the simulations we performed.

## 7 References

“California Fire Perimeters (All).” Hub-Calfire-Forestry.hub.arcgis.com, hub-calfire-forestry.hub.arcgis.com/maps/e3802d2abf8741a187e73a9db49d68fe/about. Accessed 29 Feb. 2024.

Western Fire Chiefs Association. “When and Where Do Wildfires Occur?” WFCA, 4 Mar. 2024, wfca.com/wildfire-articles/where-do-wildfires-occur/.

# Stat 250 Final Simulations

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2024-03-10

These simulations form the foundation of our CI claims and usage for our analysis of California wildfires. In the following block, we perform our simulations and print the results. Following that, we analyze the wildfire data set, determining our  $\alpha$  point estimate and 95% CI.

## Main Simulation

```
set.seed(31415) # setting seed for reproducible results.
y_m <- 5e4
alphas <- c(0.5, 1.5, 2.5) # true values of alpha.
num_alphas <- length(alphas)

ns <- c(15, 50, 100, 1000) # sample sizes.
num_ns <- length(ns)

numIter <- 1000 # number of simulations.
B <- 1000 # number of bootstrap iterations per simulation.

# set up arrays to store the estimates from our simulation
alpha_hat_MLE <- array(dim = c(num_alphas, num_ns, numIter)) # alpha mle estimators.
alpha_hat_MOM <- array(dim = c(num_alphas, num_ns, numIter)) # alpha mom estimators.
rmse_MLE <- array(dim = c(num_alphas, num_ns)) # root mean squared errors of estimators.
rmse_MOM <- array(dim = c(num_alphas, num_ns))
# exact pivotal method CI info.
exact.lower <- array(dim = c(num_alphas, num_ns, numIter))
exact.upper <- array(dim = c(num_alphas, num_ns, numIter))
exact.width <- array(dim = c(num_alphas, num_ns, numIter))
# formula t method CI info.
CLT.lower <- array(dim = c(num_alphas, num_ns, numIter))
CLT.upper <- array(dim = c(num_alphas, num_ns, numIter))
CLT.width <- array(dim = c(num_alphas, num_ns, numIter))
# bootstrap mle method CI info.
bootMLE.lower <- array(dim = c(num_alphas, num_ns, numIter))
bootMLE.upper <- array(dim = c(num_alphas, num_ns, numIter))
bootMLE.width <- array(dim = c(num_alphas, num_ns, numIter))
# bootstrap mom method CI info.
bootMOM.lower <- array(dim = c(num_alphas, num_ns, numIter))
bootMOM.upper <- array(dim = c(num_alphas, num_ns, numIter))
bootMOM.width <- array(dim = c(num_alphas, num_ns, numIter))
# bootstrap t method CI info.
bootT.lower <- array(dim = c(num_alphas, num_ns, numIter))
bootT.upper <- array(dim = c(num_alphas, num_ns, numIter))
```

```

bootT.width <- array(dim = c(num_alphas, num_ns, numIter))
CI_coverage_exact <- array(dim = c(num_alphas, num_ns)) # CI coverage tables.
CI_coverage_CLT <- array(dim = c(num_alphas, num_ns))
CI_coverage_bootMLE <- array(dim = c(num_alphas, num_ns))
CI_coverage_bootMOM <- array(dim = c(num_alphas, num_ns))
CI_coverage_bootT <- array(dim = c(num_alphas, num_ns))
CI_width_exact <- array(dim = c(num_alphas, num_ns)) # CI width tables.
CI_width_CLT <- array(dim = c(num_alphas, num_ns))
CI_width_bootMLE <- array(dim = c(num_alphas, num_ns))
CI_width_bootMOM <- array(dim = c(num_alphas, num_ns))
CI_width_bootT <- array(dim = c(num_alphas, num_ns))

for(i in 1:num_alphas){ #i indexes the alpha value.
  for(j in 1:num_ns){ #j indexes the sample size.
    for(k in 1:numIter){ #k indexes the simulation number for a fixed n and alpha.

      ## 1. Sample n i.i.d. Pareto random variables using part (a) of derivations:
      ## (Note: there is not a built-in Pareto simulator in R, so this is the best way to
      ## simulate from the distribution.)

      X <- rexp(ns[j], rate = alphas[i])
      Y <- y_m * exp(X) # Y ~ Pareto(y_m, alpha).
      Ybar <- mean(Y) # take mean of Y.
      x_se <- sd(X) # take mean and sd of X.
      Xbar <- mean(X)

      ## 2. Compute and store the MLE and MOM for alpha:
      alpha_hat_MLE[i, j, k] <- 1/(mean(X))

      alpha_hat_MOM[i, j, k] <- Ybar/(Ybar - y_m)

      ## 3. compute 95% confidence intervals for alpha using exact and formula t methods:
      exact.lower[i, j, k] <- alpha_hat_MLE[i, j, k] * qgamma(0.025, ns[j], ns[j])
      # based on exact MLE CI (part d)
      exact.upper[i, j, k] <- alpha_hat_MLE[i, j, k] * qgamma(0.975, ns[j], ns[j])
      # based on exact MLE CI (part d)
      exact.width[i, j, k] <- exact.upper[i, j, k] - exact.lower[i, j, k]

      CLT.lower[i, j, k] <- 1/(Xbar - qt(0.025, df = ns[j]-1)*x_se/sqrt(ns[j]))
      # based on t distribution CI (part e)
      CLT.upper[i, j, k] <- 1/(Xbar - qt(0.975, df = ns[j]-1)*x_se/sqrt(ns[j]))
      # based on t distribution CI (part e)
      CLT.width[i, j, k] <- CLT.upper[i, j, k] - CLT.lower[i, j, k]

      # define arrays to store bootstrap distributions.
      bootMLE <- numeric(B)
      bootMOM <- numeric(B)
      bootT <- numeric(B)
      for(b in 1:B){
        bootY <- sample(Y, replace = TRUE) # take bootstrap sample.
        bootX <- log(bootY/y_m) # transform bootsamp to get exponential data.
        # calculate
        bootMLE[b] <- 1/(mean(bootX))
      }
    }
  }
}

```

```

bootMOM[b] <- mean(bootY)/(mean(bootY) - y_m)
bootT[b] <- (mean(bootX) - (1/alpha_hat_MLE[i, j, k]))/(sd(bootX)/sqrt(ns[j]))
}
# bootstrap interval applied to the MLE.
bootMLE.lower[i, j, k] <- quantile(bootMLE, 0.025)
bootMLE.upper[i, j, k] <- quantile(bootMLE, 0.975)
bootMLE.width[i, j, k] <- bootMLE.upper[i, j, k] - bootMLE.lower[i, j, k]

# bootstrap interval applied to the MOM.
bootMOM.lower[i, j, k] <- quantile(bootMOM, 0.025)
bootMOM.upper[i, j, k] <- quantile(bootMOM, 0.975)
bootMOM.width[i, j, k] <- bootMOM.upper[i, j, k] - bootMOM.lower[i, j, k]

# bootstrap interval applied to the t-statistic.
bootT.lower[i, j, k] <- 1/(Xbar - quantile(bootT, 0.025)*x_se/sqrt(ns[j]))
bootT.upper[i, j, k] <- 1/(Xbar - quantile(bootT, 0.975)*x_se/sqrt(ns[j]))
bootT.width[i, j, k] <- bootT.upper[i, j, k] - bootT.lower[i, j, k]

} # end loop over simulation number.

## 6. Compute the RMSEs for the MLE and MOM estimates.
rmse_MLE[i, j] <- sqrt(mean((alpha_hat_MLE[i, j, ] - alphas[i])^2))
rmse_MOM[i, j] <- sqrt(mean((alpha_hat_MOM[i, j, ] - alphas[i])^2))

## 7. Compute the widths and coverage rates for our CIs.
# exact pivotal method.
CI_coverage_exact[i, j] <- mean((alphas[i] >= exact.lower[i, j, ]) &
(alphas[i] <= exact.upper[i, j, ]))
CI_width_exact[i, j] <- mean(exact.width[i, j, ])

# formula t method.
CI_coverage_CLT[i, j] <- mean((alphas[i] >= CLT.lower[i, j, ]) &
(alphas[i] <= CLT.upper[i, j, ]))
CI_width_CLT[i, j] <- mean(CLT.width[i, j, ])

# bootstrap mle method.
CI_coverage_bootMLE[i, j] <- mean((alphas[i] >= bootMLE.lower[i, j, ]) &
(alphas[i] <= bootMLE.upper[i, j, ]))
CI_width_bootMLE[i, j] <- mean(bootMLE.width[i, j, ])

# bootstrap mom method.
CI_coverage_bootMOM[i, j] <- mean((alphas[i] >= bootMOM.lower[i, j, ]) &
(alphas[i] <= bootMOM.upper[i, j, ]))
CI_width_bootMOM[i, j] <- mean(bootMOM.width[i, j, ])

# bootstrap t method.
CI_coverage_bootT[i, j] <- mean((alphas[i] >= bootT.lower[i, j, ]) &
(alphas[i] <= bootT.upper[i, j, ]))
CI_width_bootT[i, j] <- mean(bootT.width[i, j, ])
}
}

```

## Simulation Results

```
# Print Results.
print("Exact Pivotal Method CI coverage and average width:")

## [1] "Exact Pivotal Method CI coverage and average width:"
CI_coverage_exact

##          [,1] [,2] [,3] [,4]
## [1,] 0.946 0.950 0.954 0.945
## [2,] 0.951 0.945 0.952 0.945
## [3,] 0.959 0.958 0.947 0.955
CI_width_exact

##          [,1]      [,2]      [,3]      [,4]
## [1,] 0.5387451 0.2824583 0.1970980 0.06190964
## [2,] 1.6112313 0.8483980 0.5917289 0.18593829
## [3,] 2.6583407 1.4192920 0.9892573 0.31011667
print("CLT Method CI coverage and average width:")

## [1] "CLT Method CI coverage and average width:"
CI_coverage_CLT

##          [,1] [,2] [,3] [,4]
## [1,] 0.907 0.934 0.952 0.946
## [2,] 0.914 0.935 0.939 0.947
## [3,] 0.925 0.944 0.937 0.947
CI_width_CLT

##          [,1]      [,2]      [,3]      [,4]
## [1,] 0.7973646 0.3120337 0.2051813 0.06208571
## [2,] 1.9536869 0.9378515 0.6174201 0.18678217
## [3,] 4.4969785 1.5566838 1.0407511 0.31127976
print("Bootstrap-MLE Method CI coverage and average width:")

## [1] "Bootstrap-MLE Method CI coverage and average width:"
CI_coverage_bootMLE

##          [,1] [,2] [,3] [,4]
## [1,] 0.892 0.921 0.946 0.941
## [2,] 0.892 0.919 0.937 0.947
## [3,] 0.903 0.941 0.932 0.939
CI_width_bootMLE

##          [,1]      [,2]      [,3]      [,4]
## [1,] 0.5733741 0.2860158 0.1966973 0.06154227
## [2,] 1.7261106 0.8605009 0.5911639 0.18526722
## [3,] 2.8384083 1.4317816 0.9964222 0.30923097
print("Bootstrap-MOM Method CI coverage and average width:")

## [1] "Bootstrap-MOM Method CI coverage and average width:"
```

```
CI_coverage_bootMOM
```

```
##      [,1] [,2] [,3] [,4]
## [1,] 0.000 0.000 0.000 0.000
## [2,] 0.643 0.668 0.726 0.783
## [3,] 0.825 0.875 0.908 0.918
```

```
CI_width_bootMOM
```

```
##      [,1]      [,2]      [,3]      [,4]
## [1,] 0.2610034 0.07054888 0.03401828 0.003404919
## [2,] 1.5064294 0.76393696 0.55397468 0.243746906
## [3,] 2.6523400 1.37862260 0.99151412 0.352679284
```

```
print("Bootstrap-T Method CI coverage and average width:")
```

```
## [1] "Bootstrap-T Method CI coverage and average width:"
```

```
CI_coverage_bootT
```

```
##      [,1] [,2] [,3] [,4]
## [1,] 0.941 0.940 0.949 0.944
## [2,] 0.948 0.947 0.949 0.946
## [3,] 0.954 0.954 0.946 0.951
```

```
CI_width_bootT
```

```
##      [,1]      [,2]      [,3]      [,4]
## [1,] 0.6401755 0.2943614 0.1994599 0.06162656
## [2,] 1.9306202 0.8867963 0.5997383 0.18552641
## [3,] 3.1674023 1.4729886 1.0113618 0.30961055
```

```
print("Root Mean Squared Errors for MLE Estimate:")
```

```
## [1] "Root Mean Squared Errors for MLE Estimate:"
```

```
rmse_MLE
```

```
##      [,1]      [,2]      [,3]      [,4]
## [1,] 0.1535254 0.07632826 0.04984757 0.01578586
## [2,] 0.4641481 0.22710691 0.15762735 0.04680478
## [3,] 0.7258023 0.36829582 0.25982751 0.07979124
```

```
print("Root Mean Squared Errors for MOM Estimate:")
```

```
## [1] "Root Mean Squared Errors for MOM Estimate:"
```

```
rmse_MOM
```

```
##      [,1]      [,2]      [,3]      [,4]
## [1,] 0.5514749 0.5133303 0.5066321 0.50064740
## [2,] 0.5237919 0.2774353 0.2061298 0.09055839
## [3,] 0.7603381 0.3952523 0.2864095 0.10164861
```

## Wildfire Data Analysis

```
# load in libraries and data, plotting it.
library(EnvStats)
```

```
## Warning: package 'EnvStats' was built under R version 4.3.3
```

```
##
```

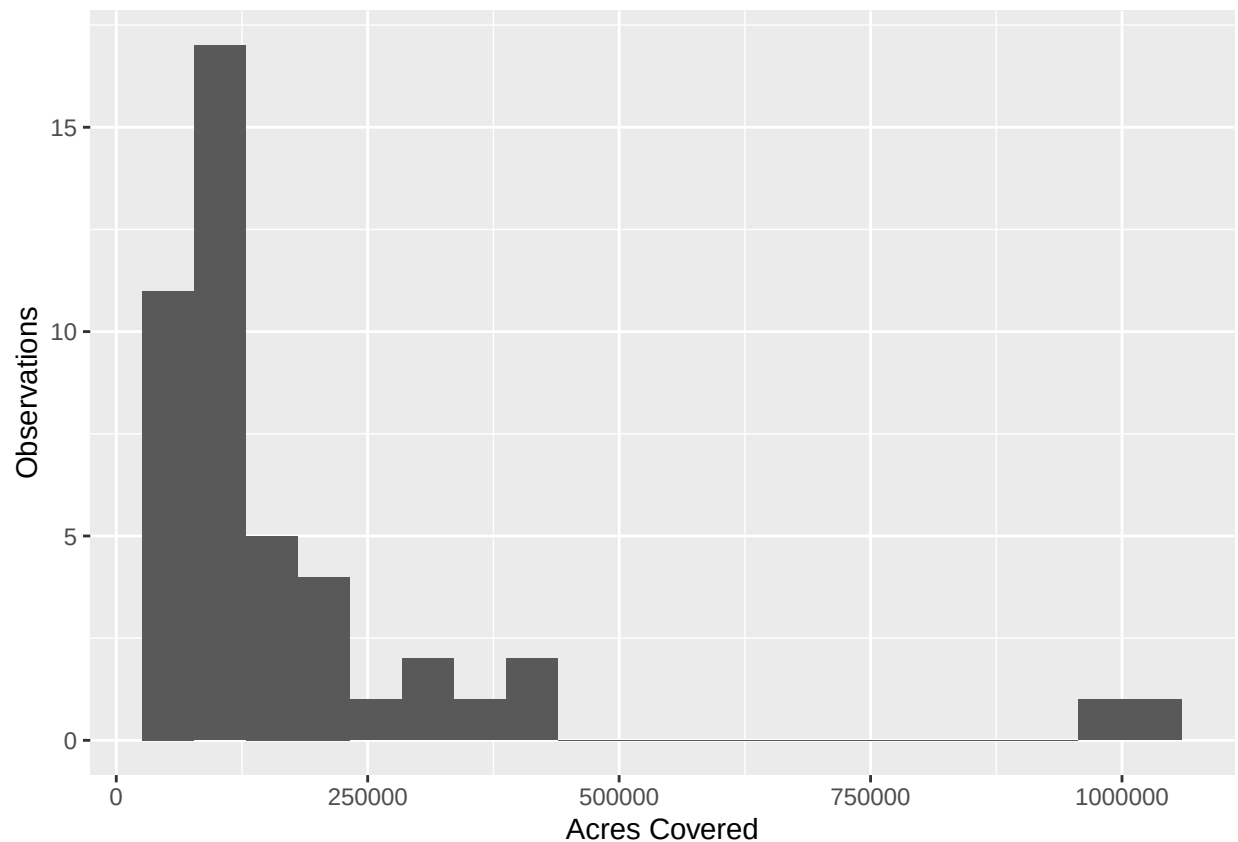
```
## Attaching package: 'EnvStats'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## predict, predict.lm
```

```
fires <- read.csv('https://people.carleton.edu/~apoppick/ClassData/CA_fires', header = TRUE)
library(ggplot2)
ggplot(NULL, aes(x = fires$GIS_ACRES)) + geom_histogram(bins = 20) +
  xlab("Acres Covered") + ylab("Observations")
```

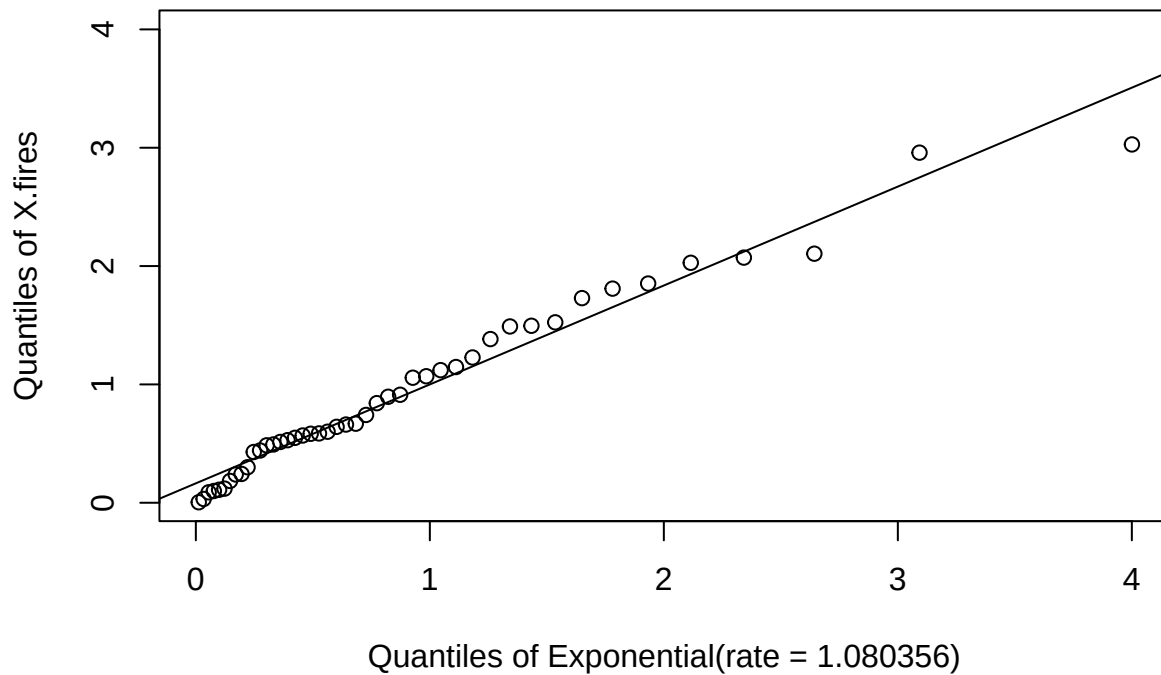


```
#define our data set, y_m, the transformed (theoretically) exponential data, and n
y_m.fires <- 50000
Y.fires <- fires$GIS_ACRES
X.fires <- log(Y.fires/y_m.fires)
n.fires <- length(Y.fires)
```

```
# make qq plot of transformed data against exponential distribution
# using the mle estimate for alpha.
```

```
qqPlot(X.fires, distribution = "exp", estimate.params=TRUE, add.line=TRUE)
```

## Exponential Q-Q Plot for X.fires



```
# calculate mle estimate for alpha, as well as exact pivotal method 95% CI.
alpha_hat.fires <- 1/mean(X.fires)
exact.lower.fires <- alpha_hat.fires * qgamma(0.025, n.fires, n.fires)
exact.upper.fires <- alpha_hat.fires * qgamma(0.975, n.fires, n.fires)

# print results.
print("Exact Pivotal Method 95% CI:")

## [1] "Exact Pivotal Method 95% CI:"
exact.lower.fires

## [1] 0.7880188
exact.upper.fires

## [1] 1.418098
print("MLE Point Estimate for Alpha:")

## [1] "MLE Point Estimate for Alpha:"
alpha_hat.fires

## [1] 1.080356
```